

Assignment 01

$$\begin{aligned}
 01. (a) \quad & (10110010101)_{25} \\
 & \quad \quad \quad -1 \\
 & \hline
 & (10110010100)_{15} \\
 & = -(01001101011)_2 \\
 & = -(2^0 + 2^1 + 2^3 + 2^5 + 2^6 + 2^9) \\
 & = -(619)_{10}
 \end{aligned}$$

$$\begin{array}{r}
 7 \overline{) 619} \\
 7 \overline{) 88-3} \\
 7 \overline{) 12-4} \\
 7 \overline{) 1-5} \\
 0-1
 \end{array}
 \uparrow$$

$$-(619)_{10} = -(1543)_7$$

$$\begin{aligned}
 (b) \quad & (00110010101)_{15} = (00110010101)_2 \\
 & = 2^0 + 2^2 + 2^4 + 2^7 + 2^8 \\
 & = (405)_{10}
 \end{aligned}$$

$$\begin{array}{r}
 9 \overline{) 405} \\
 9 \overline{) 45-0} \\
 9 \overline{) 5-0} \\
 0-5
 \end{array}
 \uparrow$$

$$(405)_{10} = (500)_9$$

$$(c) \quad (0011)_{BCD} = (3)_{10}$$

$$\begin{array}{r}
 5 \overline{) 3} \\
 0-3
 \end{array}
 \uparrow$$

$$(3)_{10} = (3)_5$$

$$\begin{aligned}
 (d) \quad & (01001010101111)_{25} = (01001010101111)_{15} \\
 & = (01001010101111)_2 \\
 & = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^6 + 2^8 + 2^{10} + 2^{13} \\
 & = (9567)_{10}
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad & (101010101000011)_{25} \\
 & \quad \quad \quad -1 \\
 & \hline
 & (101010101000010)_{15} \\
 & = -(01010101011101)_2 \\
 & = -(2^0 + 2^2 + 2^3 + 2^4 + 2^5 + 2^7 + 2^9 + 2^{11} + 2^{13})_{10} \\
 & = -(10941)_{10}
 \end{aligned}$$

$$(f) (01001010101111)_2 = (01001010101111)_2$$

$$= 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^6 + 2^8 + 2^{10} + 2^{13}$$

$$= (9567)_{10}$$

$$(g) (101010101000011)_2 = -(010101010111100)_2$$

$$= -(2^2 + 2^3 + 2^4 + 2^5 + 2^7 + 2^9 + 2^{11} + 2^{13})$$

$$= -(10940)_{10}$$

$$(h) (1100011)_{10}$$

1	1	0	0	0	1	1
+5	+5	+5	+5	+5	+5	+5
6	6	5	5	5	6	6
↓	↓	↓	↓	↓	↓	↓
(0110	0110	0101	0101	0101	0110	0110)
Ex-5						

$$02. -6 - 5 = (-6) + (-5)$$

$$-6 = (\underline{1}010)_{25}$$

$$(+)-5 = (\underline{1}011)_{25}$$

$$\begin{array}{r} (10101)_{25} \\ \hline \end{array}$$

There is overflow as we are adding 2 negative values,
but the answer is a positive number.

Canny out ←

Discarded

$$= (\underline{0}101)_{25}$$

$$03. \text{ Largest number} = +(2^{n-1} - 1) \quad n=12$$

$$= +(2^{12-1} - 1)$$

$$= +(2047)_{10}$$

$$= +(3777)_8$$

8		2047	
8		255-7	↑
8		31-7	
8		3-7	
		0-3	

$$04. \text{ Smallest number} = -(2^{n-1} - 1) \quad n=8$$

$$= -(2^{8-1} - 1)$$

$$= -(127)_{10}$$

$$= -(7F)_{16}$$

16		127	
16		7 - 15 (F)	↑
		0 - 7	

$$\begin{array}{r}
 05. \quad 124 \overline{) 43234212} \quad 00300222 \\
 \underline{432} \\
 0342^{+5} \\
 \underline{- 303^{+1}} \\
 341^{+5} \\
 \underline{- 303^{+1}} \\
 332^{+5} \\
 \underline{- 303^{+1}} \\
 24
 \end{array}$$

$$\text{Quotient} = (300222)_5$$

$$\text{Remainder} = (24)_5$$

$$124 \times 0 = 0$$

$$124 \times 1 = 124$$

$$124 \times 2 = 303$$

$$124 \times 3 = 432$$

$$124 \times 4 = 1111$$

$$\begin{aligned}
 06. \quad \text{RAM} &= (1C2)_{16} \\
 &= 2 \times 16^0 + 12 \times 16^1 + 1 \times 16^2 \\
 &= (450)_{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{Graphics Card} &= (10010110000)_2 \\
 &= 2^4 + 2^5 + 2^7 + 2^{10} \\
 &= (1200)_{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total Cost} &= 2 \times (450)_{10} + (1200)_{10} \\
 &= (2100)_{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{Borrowed} &= (4064)_8 \\
 &= 4 \times 8^0 + 6 \times 8^1 + 4 \times 8^3 \\
 &= (2100)_{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{Money Left} &= \text{Borrowed} - \text{total Cost} \\
 &= (2100)_{10} - (2100)_{10} \\
 &= (0)_{10}
 \end{aligned}$$